Are Latent Reasoning Tokens Unnecessary for Model Performance?

Anonymous ACL submission

Abstract

Training large language models to perform latent reasoning may improve performance, but the lack of legible reasoning poses challenges for model interpretability and safety assurance. This challenge has motivated interpretability research that aims to characterise the functional role of latent reasoning tokens. Some studies indicate that latent reasoning tokens are effective, whereas other work suggests that these tokens are under-utilised. In this paper, we consider the tension between existing findings by reproducing prior work which shows that some latent reasoning tokens are not necessary. Two interesting observations emerge from our experiments. First, we find that latent reasoning models often generate redundant tokens due to the fixed token budgets. Second, we observe that explicit reasoning models fine-tuned on gold reasoning traces may copy the result of the most recent reasoning step when generating an answer.

1 Introduction

Large language models (LLMs) have become increasingly central to AI research as their capabilities have been improved by scale (Kaplan et al., 2020). These capabilities are especially pronounced in explicit reasoning models (ERMs), such as OpenAI o1 (Jaech et al., 2024) and DeepSeek R1 (Guo et al., 2025), that learn to produce higher quality outputs by starting their responses with a chain-of-thought (CoT): a passage of text in which the model's given task is broken down into smaller steps that can be tackled one at a time (Wei et al., 2022). Latent reasoning models (LRMs), such as COCONUT (Hao et al., 2024) and CODI (Shen et al., 2025), replace these human-readable steps with an illegible array of numbers, as shown in Figure 1. This modification may improve reasoning performance, but it also raises new challenges for our understanding of these systems.

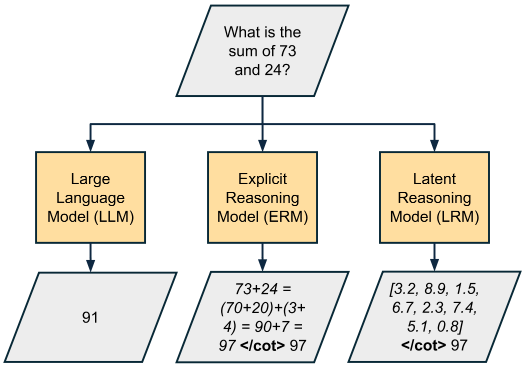

Figure 1: **A comparison of model outputs.**

Interpreting LLMs is difficult due to complications like polysemanticity and superposition (Olah et al., 2020; Elhage et al., 2022). However, explicit CoTs have proved advantageous, as they can provide observers with information about model reasoning. This information can be leveraged to improve model safeguards, such as monitoring systems, as shown by Baker et al. (2025). Korbak et al. (2025) warn that these advantages could be lost in the transition to latent reasoning.

Work on interpreting latent reasoning models has been motivated by a need to address these risks. Some research has attempted to decode latent reasoning tokens (Shen et al., 2025), or replace them with a legible representations (Wang et al., 2026). Other studies attempt to characterise the functional role of latent reasoning tokens. Hao et al. (2024) show that COCONUT uses its latent CoT to attempt multiple solutions in parallel, while Goyal et al. (2025) show that CODI's confidence in the correct answer gradually increases as additional latent reasoning tokens are generated. In contrast, Dilgren and Wiegreffe (2026) show that some LRMs may determine their final answer before generating part or all of their CoT. In this paper, we aim to address the tension between these findings.

The rest of the paper is structured as follows, in

Section 3 we replicate the experiment performed by Dilgren and Wiegreffe (2026). We proceed by performing ablations to understand the factors contributing to the discrepancy between the results reported for COCONUT, CODI, and the ERM. We identify two factors that contribute to this discrepancy: the first is the fixed token budgets used by the LRMs (Section 4), which force them to generate redundant latent reasoning tokens; the second is the tendency of the ERM to copy the result of the most recent explicit reasoning step when generating an answer (Section 5), which inflates perceived reasoning step necessity since truncating the CoT often causes the ERM to change its answer.

2 Background

Latent Reasoning Models: Latent reasoning models work by applying recurrent computations to the final hidden state. Two distinct categories of latent reasoning models can be identified: vertical recurrence models, such as Huginn-0125 (Geiping et al., 2025) and Ouro (Zhu et al., 2025), that repeatedly apply intermediate layers in a deep neural network to the final hidden state; and horizontal recurrence models, such as Switch (Yang et al., 2026) and SIM-CoT (Wei et al., 2025), that generate latent reasoning through an autoregressive process where the final hidden state is combined with the prompt and used as input to another forward pass. In this paper, we focus on two horizontal-recurrence-based latent reasoning models: COCONUT (Hao et al., 2024) and CODI (Shen et al., 2025).

Latent Reasoning Interpretability: Existing work has commented on whether or not latent reasoning tokens are necessary for model performance, but conclusions appear to be mixed. Wang et al. (2026) show that features in sparse latent reasoning tokens are causally effective operators. Li et al. (2026) find that some latent reasoning tokens have an outsized causal influence on the model's final answer, while some aren't as effective, and that a correct answer can often be decoded early. Similarly, Aswal et al. (2026) also show that the causal effect of latent reasoning tokens on the model's final answer is varied, and Chang et al. (2026) find that early latent reasoning tokens act as critical causal hubs. Goyal et al. (2025) investigate a mechanism that may explain the heterogeneity in CODI. It is shown that interventions on even latent reasoning tokens, which appear to store information, are more impactful than interventions on odd latent reasoning tokens, which appear to be transitional states produced during computation over that stored information. Cui et al. (2026) report pervasive "shortcut" behaviour, where LRMs perform well without utilising their latent reasoning tokens. Viveiros et al. (2026) examines visual LRMs and produce similar results, finding that performance wasn't significantly different when reasoning tokens were swapped out for dummy tokens. Jin et al. (2026) show that the necessity of latent reasoning tokens can change during the course of training. Our work focuses particularly on Dilgren and Wiegreffe (2026), who find that truncating the latent CoT often doesn't change the models answer, and conclude that latent reasoning tokens may not be necessary for model performance.

3 Reproduction

We reproduce the experimental setup described by Dilgren and Wiegreffe (2026), which aims to determine the extent to which reasoning tokens are necessary for model performance. Problems are sampled from a dataset and given to a reasoning model, which generates a response to each problem. A model's response is comprised of a CoT, which consists of reasoning tokens, and the model's final answer. Several truncated versions of the CoT are created by removing all reasoning tokens after a given point. The model is prefilled with the problem statement and each truncated CoT, and then forced to generate an answer as we terminate the CoT by inserting an `<eot>` token. The resulting answers are compared to the model's original answer to determine the impact of the removed reasoning. In the rest of this section we provide additional details on the experimental setup.

Datasets: We use the same 3 datasets as Dilgren and Wiegreffe (2026): GSM8k (Cobbe et al., 2021), ProsQA (Hao et al., 2024), and PrOntoQA (Saparov and He, 2022). GSM8k consists of numerical math problems solvable by "a bright middle school student", whereas both ProsQA and PrOntoQA consist of logic problems. All instances from each dataset include a problem statement, an answer, and an exemplar reasoning trace. Examples are provided in Appendix A. Each dataset is split into a training partition, which is used to fine-tune our models, and a testing partition. More details on dataset partitions are provided in Appendix B.

Model	GSM8k		ProsQA		PrOntoQA	
	Ours	**D&W**	**Ours**	**D&W**	**Ours**	**D&W**
ERM	41.5%	41.6%	74.2%	74.2%	99.2%	99.3%
COCONUT	33.1%	33.1%	93.8%	98.0%	98.9%	99.0%
CODI	41.6%	42.2%	81.4%	81.6%	94.9%	95.1%

Table 1: **Testing accuracy of explicit reasoning model and two latent reasoning models (COCONUT, CODI) on three datasets.** D & W defines the corresponding score reported by Dilgren and Wiegreffe (2026).

Models: We evaluate the same three models used by Dilgren and Wiegreffe (2026), all of which are trained by fine-tuning GPT-2 small (Radford et al., 2019). The first model is an ERM trained by fine-tuning on exemplar reasoning traces. The second and third models have latent reasoning capabilities, and are trained by applying the COCONUT (Hao et al., 2024) and CODI (Shen et al., 2025) training procedures. We do not train these models ourselves, but instead use the checkpoints provided by the original authors.[1] In Table 1 we show that these checkpoints produce similar accuracy scores as those reported by Dilgren and Wiegreffe (2026).

Metrics: We compute the two metrics described by Dilgren and Wiegreffe (2026). The first is *first match*, which is equal to the minimum number of reasoning tokens that must be generated before a model first produces the same answer as it would have if it had finished generating its CoT. The second metric is *stable match*, which is equal to the minimum number of reasoning tokens after which the model does not change its answer regardless of the number of additional tokens remaining. In line with Dilgren and Wiegreffe (2026), we divide first match and stable match by the number of latent reasoning tokens or explicit reasoning steps in the corresponding CoT and report the fraction as the result. In Figure 2 we show that our reproduction achieves similar matching metrics as Dilgren and Wiegreffe (2026), we expect that differences are the result of run-to-run variation.

4 Fixed Token Budgets Produce Redundant Reasoning

Dilgren and Wiegreffe (2026) propose two hypotheses that may explain why COCONUT and CODI produce lower matching metrics than the ERM: the first is that low task difficulty may cause matching metrics to be low in COCONUT and CODI, but not the ERM; the second is that COCONUT and CODI under-utilise their reasoning tokens relative to the ERM. Our results in this section support the first hypothesis, since we find that the LRM's matching metrics are affected substantially by task difficulty, *i.e.,* they are lower on less difficult tasks, while the ERM's matching metrics are not, as shown in Figure 3. We proceed by refining the second hypothesis: we suggest that COCONUT and CODI may generate a similar number of effective reasoning steps as the ERM, as shown in Figure 4, but they also generate additional redundant reasoning steps on simple questions to fill their fixed reasoning token budget.

Task Difficulty Is Associated With Stable Match We bucket GSM8k samples by the number of computational steps represented with calculator notation in the gold reasoning trace. Figure 3 shows that as the number of steps in a sample's gold reasoning trace increases, the accuracy of all three models decreases – validating step count as a proxy for task difficulty. On the other hand LRMs' stable match fraction tends to increase, while the ERM's stable match fraction remains close to saturated.

Fixed Budgets Force Redundant Tokens Figure 4 shows that all three models generate a similar number of effective reasoning steps, which means that lower stable match scores in the LRMs are the result of additional redundant reasoning tokens. We suggest that the LRMs generate these redundant tokens because, unlike the ERM, they have a fixed reasoning token budget. Simple questions may require fewer tokens than the budget permits, and residual tokens may not affect the model's answer.

ProsQA and PrOntoQA We repeat this analysis on ProsQA (Appendix C); PrOntoQA is excluded because every exemplar trace contains exactly six steps, yielding a single bucket. ProsQA proves to be a degenerate case for this analysis: accuracy is flat across buckets for all three models, so gold step count does not track difficulty on this dataset,

[1] https://huggingface.co/collections/connordilgren/are-latent-reasoning-models-easily-interpretable, https://huggingface.co/zen-E/CODI-gpt2

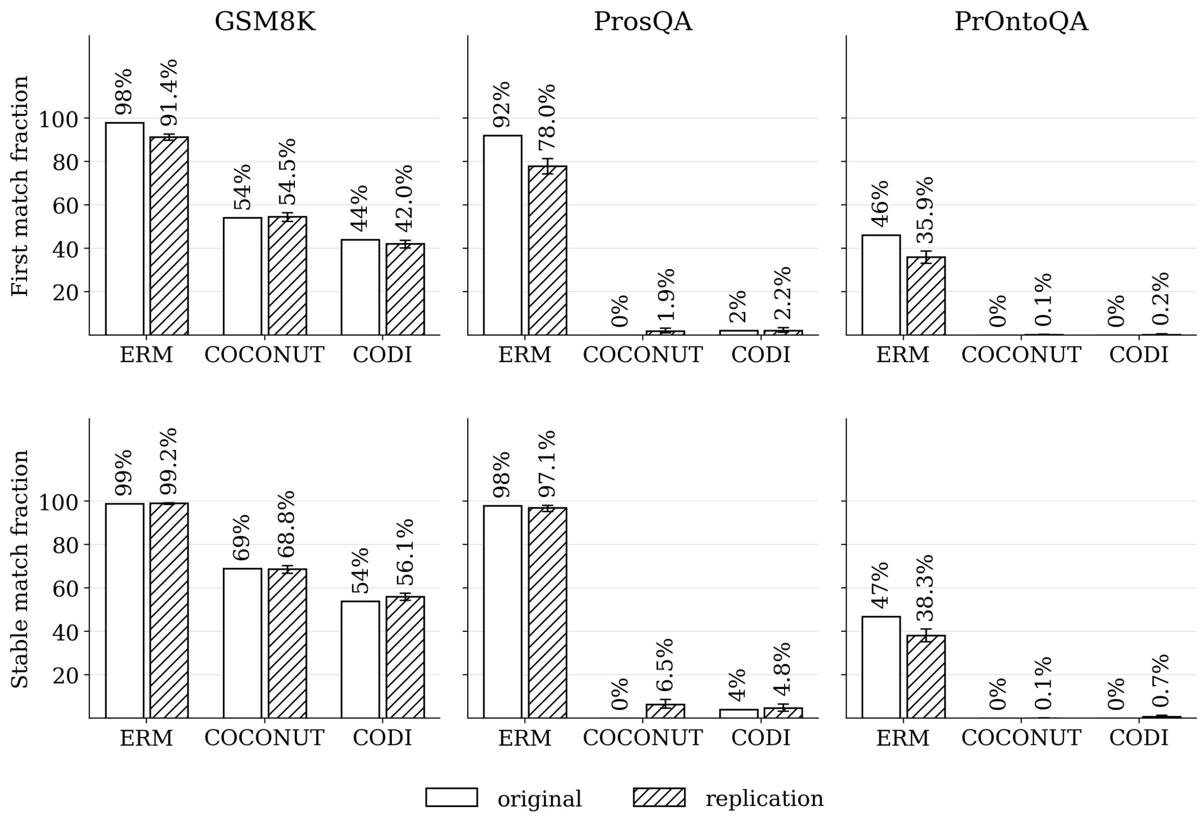

Figure 2: **Our reproduction produces similar *first match* and *stable match* scores as Dilgren and Wiegreffe (2026).** Error bars on the reproduction bars are a 95% bias-corrected and accelerated (BCa) bootstrap confidence interval (10,000 resamples) over the per-question first/stable match fractions; no comparable interval is available for the originally reported numbers, since only a single summary figure was published for each.

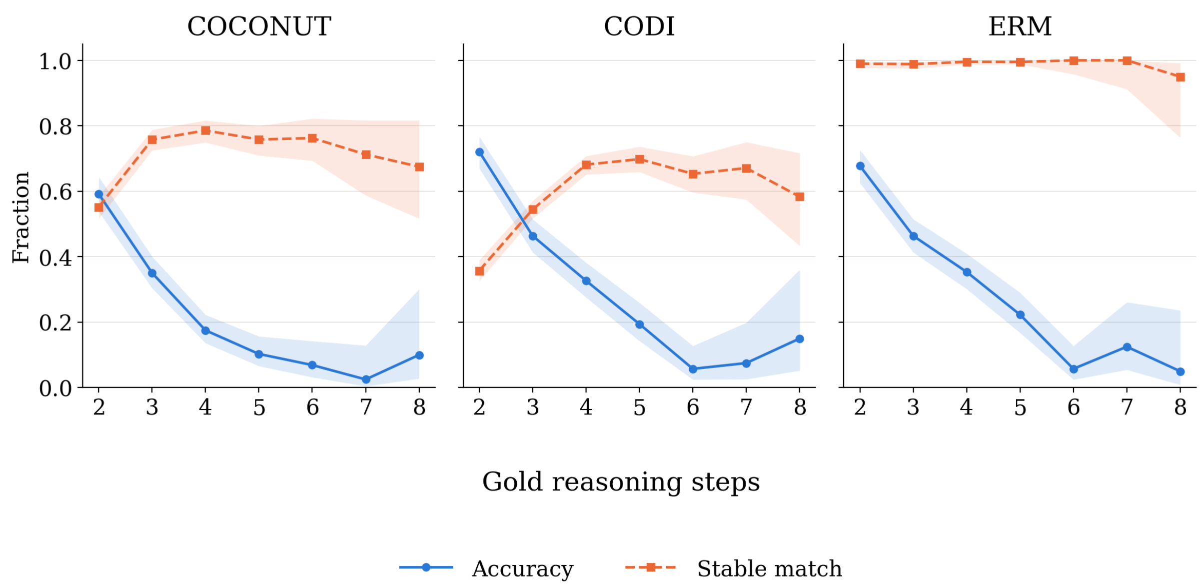

Figure 3: **Testing accuracy and stable match score with increasing gold reasoning steps (GSM8k).** Shaded region is 95% confidence interval: a Wilson score interval for accuracy, since it is a binomial correct/incorrect proportion; and a BCa bootstrap confidence interval (10,000 resamples) over per-question stable match fractions for stable match. As number of steps increases task difficulty is increased, as shown by reduced performance, and, in COCONUT and CODI, stable match increases.

and the LRMs' stable match sits near zero in every bucket – consistent with the fixed-budget account, under which uniformly easy problems render the whole latent budget redundant. Correspondingly, stable match diverges across models on ProsQA: the LRMs' falls near zero, while the ERM's approaches its full CoT length, which Section 5 suggests partly reflects its copying bias rather than additional computation.

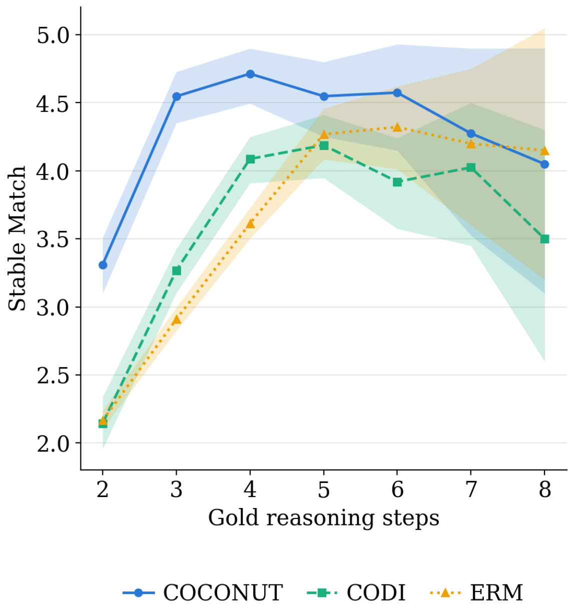

Figure 4: **Effective reasoning steps of three models with increasing gold reasoning steps (GSM8k).** All three models emit a similar number of effective reasoning steps on GSM8k questions with a given number of steps in the gold reasoning trace. Shaded regions are a 95% BCa bootstrap confidence interval (10,000 resamples) over per-question effective-step values.

5 ERM's Answers May Be Copied From the Last Reasoning Step

In this section we investigate whether the ERM has a tendency to copy the result of the last reasoning step when generating answers. This behaviour could account for some of the discrepancy in matching metrics between the LRMs and the ERM, since it could cause the ERM to change its answer each time the CoT is truncated.

Experimental Setup: Each question in a given dataset is augmented. Then, each model is applied to both the original question, and the augmented variant. The resulting explicit reasoning steps and latent reasoning tokens are recorded. We proceed by running each model on the original question with truncated reasoning traces, as in Section 3. Before the model answers, we swap the last explicit reasoning step or latent reasoning token with the corresponding reasoning tokens generated for the augmented question.

Datasets: We perform this experiment on both GSM8k and ProsQA. PrOntoQA is left for future work. GSM8k questions are augmented by swapping the first number in a problem statement with a random integer within the same order of magnitude. ProsQA questions are augmented by swapping every occurrence of two words throughout the problem statement. Specifically, question endings are of the form "Is NAME a OPT1 or OPT2?". We identify which of OPT1/OPT2 is correct (*target*) and which is not (*neg_target*), then swap every occurrence of these two words throughout the question – in every "Every Y is a Z." rule as well as the query itself. Because this is a pure relabelling of two symbols, any derivation chain that previously concluded *target* now concludes *neg_target* and vice versa, so the correct answer is guaranteed to flip while every rule in the question remains logically valid.

Exclusion Criteria: We exclude samples where: 1) a different number of explicit reasoning steps were produced for the original question and the augmented question, 2) the GSM8k question did not contain a number that could be perturbed, 3) the model gave the same answer to both questions at a given degree of CoT truncation, and 4) the model's last reasoning step was not effective. More details on excluded samples is available in Appendix D.

Answer Categorisation: Answers are categorised as *augmented*, if the swap causes the model's answer to match the answer given to the augmented question, *original* if the swap resulted in no change, or *other* if the resulting answer no longer matched either the original answer or the augmented one.

Results: In Figure 5 and Figure 6 we see that the ERM changes its answer substantially more often than either LRM, suggesting that the LRMs do not have an analogous copying bias to the ERM.

6 Conclusion

In this paper, we successfully reproduce the results of Dilgren and Wiegreffe (2026), and design targeted experiments to identify two factors that may explain the gap in matching metrics between

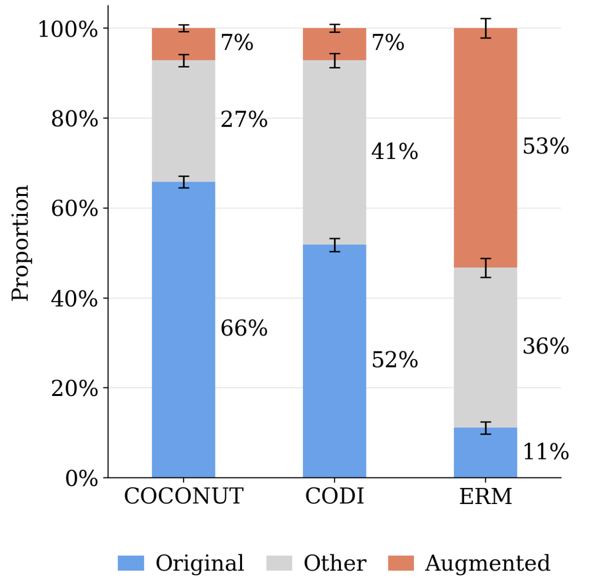

Figure 5: **Answer status after splicing one augmented-question reasoning step into an otherwise-original trace (GSM8k).** The resulting answer either still matches the original question (*original*), switches to match the augmented question (*augmented*), matches neither (*other*). The ERM switches to the augmented answer far more often than COCONUT or CODI, consistent with a copying bias. Error bars are a 95% cluster bootstrap confidence interval (10,000 resamples): whole questions, rather than individual steps, are resampled with replacement, since steps from the same question are correlated rather than independent trials.

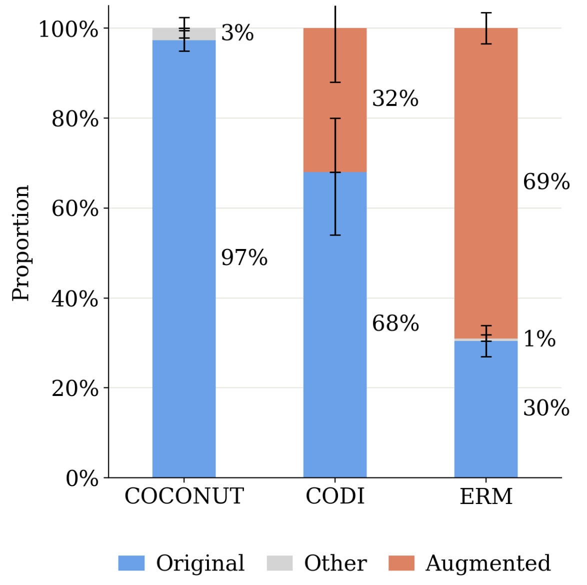

Figure 6: **Answer status after splicing one augmented-question reasoning step into an otherwise-original trace (ProsQA).** As Figure 5, but with the augmented question built by swapping the query's two named class options everywhere they appear (see text). The ERM switches to the augmented answer far more often than COCONUT or CODI, consistent with Figure 5. Error bars are a 95% cluster bootstrap confidence interval (10,000 resamples) over whole questions.

the ERM, and the LRMs (COCONUT and CODI). First, fixed token budgets force latent reasoning models to generate redundant tokens, thereby reducing their matching metrics relative to the ERM. Second, the ERM has a tendency to copy the result of the most recent explicit reasoning step when generating its final answer, thereby increasing its matching metrics relative to the LRMs. These findings refine the hypothesis that COCONUT and CODI under-utilise their latent reasoning tokens. Instead, we conclude that they effectively use tokens early in the CoT (Chang et al., 2026), the number of which varies with problem complexity. Furthermore, our results suggest that truncation-based evaluations of CoT necessity, and faithfulness tests built on the same operation (**?**), are sensitive to the way in which a model uses its CoT. We suggest that future work investigates whether our evaluations of model faithfulness can be influenced by whether the model explores multiple candidate solutions in its CoT, and by where key results appear within the model's CoT.

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A Dataset Examples

We give one representative test-split instance from each dataset used in our experiments (Section 3), chosen among the shorter instances in each dataset for space. GSM8k's gold answer is a natural-language reasoning trace with numerical operations annotated in calculator notation (Section 3); ProsQA and PrOntoQA instead provide their gold reasoning as an explicit list of steps, with a short final answer.

A.1 GSM8K

Question: An eraser costs $2 and a pencil costs $3. How much do 6 erasers and 8 pencils cost?

Gold answer:

> Six erasers cost $2/eraser x 6 erasers = $<<2*6=12>>12. Eight pencils cost $3/pencil x 8 pencils = $<<3*8=24>>24. So, 6 erasers and 8 pencils cost $12 + $24 = $36. #### 36

A.2 ProsQA

Question: Every zumpus is a timpus. Every jompus is a terpus. Eva is a sterpus. Eva is a zumpus. Max is a rorpus. Every timpus is a lorpus. Every jompus is a rorpus. Every tumpus is a terpus. Every sterpus is a zumpus. Every terpus is a gerpus. Tom is a storpus. Every zumpus is a jompus. Every sterpus is a timpus. Every storpus is a boompus. Tom is a boompus. Every timpus is a tumpus. Is Eva a boompus or gerpus?

Gold reasoning steps:

1. Eva is a zumpus.
2. Every zumpus is a jompus.
3. Every jompus is a terpus.
4. Every terpus is a gerpus.

Answer: Eva is a gerpus.

A.3 PrOntoQA

Question: Grimpuses are angry. Jompuses are lempuses. Zumpuses are earthy. Grimpuses are jompuses. Every gorpus is not hot. Jompuses are moderate. Zumpuses are dumpuses. Each lorpus is not spicy. Shumpuses are dull. Lempuses are yumpuses. Tumpuses are sterpuses. Wumpuses are shy. Lempuses are wumpuses. Each yumpus is opaque. Yumpuses are tumpuses. Lempuses are not small. Tumpuses are impuses. Yumpuses are vumpuses. Tumpuses are not dull. Sterpuses are discordant. Vumpuses are brown. Grimpuses are gorpuses. Jompuses are lorpuses. Rex is a zumpus. Rex is a grimpus. True or false: Rex is dull.

Gold reasoning steps:

1. Rex is a grimpus. Grimpuses are jompuses.
2. Rex is a jompus. Jompuses are lempuses.
3. Rex is a lempus. Lempuses are yumpuses.
4. Rex is a yumpus. Yumpuses are tumpuses.
5. Rex is a tumpus. Tumpuses are not dull.
6. Rex is not dull.

Answer: False

B Dataset Splits

Table 2 gives the train/validation/test split sizes for each dataset used in our experiments (Section 3). All of our own evaluation is run on the test split only, using models already fine-tuned by the original authors; we did not do any training ourselves. GSM8K's splits come from its standard Hugging Face Hub mirror (`openai/gsm8k`), which does not define a validation split. ProsQA and PrOntoQA's splits come from the replicated paper's own repository (Section 3).

Dataset	Train	Valid	Test
GSM8K	7,473	—	1,319
ProsQA	17,886	300	500
PrOntoQA	9,000	200	800

Table 2: Number of instances in each split, by dataset.

C Bucketing ProsQA by Steps

Section 4 repeats the GSM8k gold-step-count bucketing analysis (Figure 3) on ProsQA, bucketing each test sample by the length of its gold reasoning chain (Section A) rather than by calculator-notation steps. Figure 7 shows the result: unlike on GSM8k, accuracy does not decrease as gold step count increases for any of the three models, so step count is not a difficulty signal on this dataset. COCONUT and CODI's stable match stays near zero across every bucket, while the ERM's stays near its maximum – the same divergence seen on GSM8k, but without the step-count trend that on GSM8k ties it to task difficulty.

D Sample Exclusion for the Copying-Bias Experiment

Section 5 excludes some GSM8k test samples from the copying-bias slicing experiment (Figure 5) before computing each model's *original*, *augmented*, and *other* proportions. Table 3 shows how many of the 1,319 GSM8k test samples remain for each model after these exclusions. A third criterion mentioned in Section 5 entails excluding cases where the model gives the same answer to the original and augmented questions at a given degree of CoT truncation. We label these cases as *Ties*, and show the remaining sample counts in Table 4. Tables 5 and 6 give the same breakdown for the ProsQA version of this experiment (Figure 6).

Model	No number	Step mismatch	Remaining
ERM	23	353	943
COCONUT	23	—	1,296
CODI	23	—	1,296

Table 3: GSM8k test samples excluded from the copying-bias slicing experiment (Figure 5), out of 1,319 total, and the number remaining. "No number": the question contains no number that can be perturbed to build an augmented question. "Step mismatch": the original and augmented questions produced a different number of explicit reasoning steps (ERM only).

Model	Before	Ties	After
ERM	2,876	640	2,236
COCONUT	7,776	1,413	6,363
CODI	7,776	802	6,974

Table 4: Splicing trials underlying Figure 5, summed over the remaining samples in Table 3: total trials before removing ties, the number of those tagged *tie*, and the number of non-tie trials remaining, which is what Figure 5's *original*/*augmented*/*other* proportions are computed over.

Model	Unparseable	Step mismatch	Remaining
ERM	0	83	417
COCONUT	0	—	500
CODI	0	—	500

Table 5: ProsQA test samples excluded from the copying-bias slicing experiment (Figure 6), out of 500 total, and the number remaining.

Model	Before	Ties	After
ERM	1,613	1,078	535
COCONUT	3,000	111	2,889
CODI	3,000	549	2,451

Table 6: Splicing trials underlying Figure 6, summed over the remaining samples in Table 5. Compared to Table 4, the ProsQA word-swap augmentation touches several rules per question rather than a single number, which leaves far fewer ties for COCONUT and CODI (and a correspondingly larger set of non-tie trials) than the GSM8k version of this experiment.

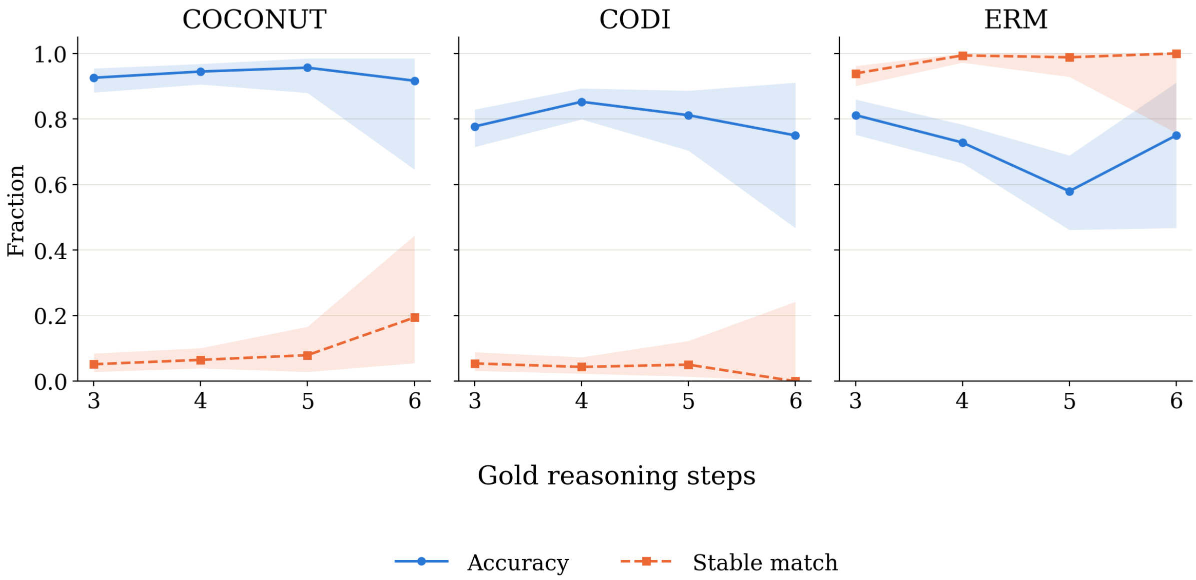

Figure 7: **Testing accuracy and stable match score with increasing gold reasoning steps (ProsQA).** As Figure 3, but bucketed by the length of each sample's gold reasoning chain instead of calculator-notation steps. Shaded region is a 95% confidence interval: a Wilson score interval for accuracy, and a BCa bootstrap confidence interval (10,000 resamples) over per-question stable match fractions for stable match. Accuracy is roughly flat across buckets for all three models, and COCONUT and CODI's stable match remains near zero throughout, in contrast to the GSM8k trend in Figure 3.